**Figure 5.1 Elements of trust and integrity**

We begin with cryptography, which is at the core of generating account addresses as an identity for the decentralized participants. Then we explore the hashing techniques to implement security and privacy. You'll learn how to apply these concepts, tools, and techniques in designing and developing smart contracts with security and privacy features.

We will explore an example application that solves a specific decentralized use case: a decentralized blind auction. Additionally, the blind auction example will reinforce the smart contract design principles you learned in chapters 2-4.

In this chapter, you'll see how to deploy smart contracts on a new tool: a public test chain called Ropsten. This approach is a first gradual step forward before deploying on a real production chain: the Ethereum mainnet, where operations cost real ether. The reason for introducing a public chain in this chapter is to highlight the importance of privacy and security when you deploy on a public chain. Let's begin with cryptography.

# 5.1 Cryptography basics

Bitcoin and its working cryptocurrency model are based on a strong foundation of cryptographic research and algorithms developed over more than 40 years. In most of your everyday programming projects, security is implicit. On the other hand, you'll find that cryptography plays an indispensable and explicit role in a decentralized blockchain-based solution, in which it is used for

- Creating a digital identity for the participants and other entities
- Securing data and transactions
- Ensuring the privacy of data
- Signing documents digitally